Molecular orbital theory describes a many-electron system in terms of one-electron spatial orbitals

$$\phi_i^\mu = \phi_i \left(x^\mu, y^\mu, z^\mu\right). \tag{1}$$

Here, ϕ_i^μ denotes the ith molecular orbital evaluated at the spatial coordinates (x^μ, y^μ, z^μ) of the μth electron. For simplicity, the nuclei are considered fixed, and magnetic effects arising from the spins and orbital motions of the electrons are neglected [1]. Assuming a closed shell system e.g. (He, Li^+, F^-), each molecular orbital has two electrons (spin up and spin down), satisfying the Pauli exclusion principle. An extension of the theory is to define molecular spin orbitals (MSO), so that only one electron can exists in each MSO. The total N-electron wavefunction is built up as a product of such MSO's.The MSO's that constitute a closed shell also form an orthonormal set [1]. The Hartree-Fock wave equation is defined [2, 3] to satisfy the antisymmetrisation postulate for fermions,

$$\Psi_{\mathrm{HF}}(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(x_1) & \psi_2(x_1) & \cdots & \psi_N(x_1) \\ \psi_1(x_2) & \psi_2(x_2) & \cdots & \psi_N(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_1(x_N) & \psi_2(x_N) & \cdots & \psi_N(x_N) \end{vmatrix}. \tag{2}$$

The electronic energy of the system is given by

$$E = \int \Psi_{\mathrm{HF}}^* \mathcal{H} \Psi_{\mathrm{HF}} \, \mathrm{d}\tau, \tag{3}$$

where the total electronic Hamiltonian is

$$\mathcal{H} = \sum_\mu \hat{H}^\mu + \frac{e^2}{2} \sum_{\mu \neq \nu} \frac{1}{r^{\mu\nu}}. \tag{4}$$

The first term contains the one-electron Hamiltonian for each electron, while the second term describes the electrostatic repulsion between the electrons. The factor of one-half prevents each electron pair from being counted twice.

For closed-shell systems [1],

$$E = 2 \sum_i H_i + \sum_{ij} \left(2J_{ij} - K_{ij}\right). \tag{5}$$

The indices i and j label occupied spatial molecular orbitals. The quantity H_i is the expectation value of the one-electron Hamiltonian for an electron occupying the ith spatial molecular orbital. The term $2\sum_i H_i$ represents the total one-electron energy, where the factor of two arises from the double occupancy of each spatial molecular orbital:

$$H_i = \int \bar{\phi}_i^\mu \, \hat{H}^\mu \phi_i^\mu \, \mathrm{d}v^\mu. \tag{6}$$

Recalling that ϕ_i^μ is the ith spatial molecular orbital ϕ_i evaluated at the coordinates of the μth electron. H^μ is the corresponding one-electron Hamiltonian:

$$H^\mu = \frac{(P^\mu)^2}{2m} - \sum_n \frac{Z_n e^2}{r^{\mu n}}. \tag{7}$$

The first term is the kinetic-energy operator of the μth electron, while the second term describes its electrostatic attraction to all nuclei.

Furthermore, J_{ij} is the Coulomb integral:

$$J_{ij} = J_{ji} = \bar{J}_{ij} = \bar{J}_{ji} = e^2 \int \frac{\bar{\phi}_i^\mu \bar{\phi}_j^\nu \phi_i^\mu \phi_j^\nu}{r^{\mu\nu}} \, \mathrm{d}v^{\mu\nu}. \tag{8}$$

The physical interpretation of J_{ij}, for $i \neq j$, is that it represents the mutual electrostatic repulsion energy of the two electron clouds associated with the molecular orbitals ϕ_i and ϕ_j. Equivalently, J_{ij} is the work required to bring the two electron clouds from infinite separation to their final positions in the atom or molecule.

Finally, K_{ij} is the exchange integral, which arises from the antisymmetry of the electronic wavefunction and accounts for the indistinguishability of fermions. It has no direct classical interpretation:

$$K_{ij} = K_{ji} = \bar{K}_{ij} = \bar{K}_{ji} = e^2 \int \frac{\bar{\phi}_i^\mu \bar{\phi}_j^\nu \phi_j^\mu \phi_i^\nu}{r^{\mu\nu}} \, \mathrm{d}v^{\mu\nu}. \tag{9}$$

Note the assumption of orthonormal molecular orbitals ϕ_i, the assumption remains pertinent throughout all future calculations. If a non-orthonormal set of molecular orbitals were used, a more complicated expression for the total electronic energy would result.

To find the Hartree–Fock ground state, we apply the variational principle within the Hartree–Fock approximation. A small variation is applied to the molecular orbitals,

$$\phi_i \to \phi_i + \delta\phi_i.$$

The variation must preserve the orbital orthonormality constraint,

$$\int \phi_i^* \phi_j \, \mathrm{d}v = \delta_{ij}. \tag{10}$$

Under this variation,

$$\delta E = 2 \sum_i \delta H_i + \sum_{ij} \left(2\delta J_{ij} - \delta K_{ij} \right). \tag{11}$$

Where,

$$2 \sum_i \delta H_i = 2 \sum_i \left[\int \delta(\phi_i^*) \hat{H} \phi_i \, \mathrm{d}v + \int \phi_i^* \hat{H} \delta(\phi_i) \, \mathrm{d}v \right]. \tag{12}$$

$$\delta J_{ij} = e^2 \int \frac{\delta(\phi_i^{*\mu})\phi_j^{*\nu}\phi_i^{\mu}\phi_j^{\nu} + \phi_i^{*\mu}\delta(\phi_j^{*\nu})\phi_i^{\mu}\phi_j^{\nu} + \phi_i^{*\mu}\phi_j^{*\nu}\delta(\phi_i^{\mu})\phi_j^{\nu} + \phi_i^{*\mu}\phi_j^{*\nu}\phi_i^{\mu}\delta(\phi_j^{\nu})}{r^{\mu\nu}} \, \mathrm{d}v^{\mu\nu}. \tag{13}$$

Consider the first term:

$$\begin{aligned} e^2 \int \frac{\delta(\phi_i^{*\mu})\phi_j^{*\nu}\phi_i^{\mu}\phi_j^{\nu}}{r^{\mu\nu}} \, \mathrm{d}v^{\mu\nu} &= \int \delta(\phi_i^{*\mu}) \underbrace{\left[e^2 \int \frac{\phi_j^{*\nu}\phi_j^{\nu}}{r^{\mu\nu}} \, \mathrm{d}v^{\nu} \right]}_{J_j} \phi_i^{\mu} \, \mathrm{d}v^{\mu}. \\ &= \int \delta(\phi_i^{*\mu}) J_j \phi_i^{\mu} dv^{\mu} \end{aligned} \tag{14}$$

Thus, we simplify

$$\delta J_{ij} = 2 \int \delta\phi_i^*(J_j)\phi_i dv + 2 \int \phi_i^*(J_j)\delta\phi_i dv$$

A similar expression can be derived for K_{ij} [1], where

$$K_j = e^2 \int \frac{\phi_j^{*\nu}\phi_j^{\nu}}{r^{\mu\nu}} dv^{\nu}. \tag{15}$$

So that,

$$\delta E = 2\sum_i \int \delta\phi_i^*[H + \sum_j (2J_j - K_j)]\phi_i dv + 2\sum_i \int \delta\phi_i[H^* + \sum_j (2J_j^* - K_j^*)]\phi_i^* dv \tag{16}$$

Now, we consider the constraint

$$\int \phi_i^*\phi_i = \delta_{ij}$$

which holds true if the derivative

$$\int \delta(\phi_i^*)\delta_j dv + \int d(\phi_j)\phi_i^* dv = 0. \tag{17}$$

Define $\epsilon_{ij} = \epsilon_{ji}$ as our lagrange multiplier, so that

$$\begin{aligned} \delta E' = 2\sum_i \int \delta\phi_i^*\{[H + \sum_j (2J_j - K_j)]\phi_i - \sum_j \phi_j \epsilon_{ji}\} dv \\ + 2\sum_i \int \delta\phi_i\{[H^* + \sum_j (2J_j^* - K_j^*)]\phi_i^* - \sum_j \phi_j^* \epsilon_{ij}\} dv \end{aligned} \tag{18}$$

To minimise the energy, we require $dE' = 0$. This results in two equations,

$$[H + \sum_j (2J_j - K_j)]\phi_i = \sum_j \phi_j \epsilon_{ji} \tag{19}$$

$$[H^* + \sum_j (2J_j^* - K_j^*)]\phi_i^* = \sum_j \phi_j^* \epsilon_{ij}. \tag{20}$$

The complex conjugate of Eq. (20) minus Eq. (19) leads to the result that

$$\sum_j \phi_j (\epsilon_{ji} - \epsilon_{ij}^*) = 0.$$

It was stipulated that the molecular orbitals ϕ_i are linearly independent and so

$$\epsilon_{ij}^* = \epsilon_{ji}. \tag{21}$$

We can conclude the ϵ matrix is Hermitian.

Finally, define

$$G = \sum_i (2J_i - K_i) \quad \text{and} \quad F = H + G. \tag{22}$$

Since ϵ is Hermitian, it can be diagonalised, thus reducing Eq. (19) and Eq. (20)to the matrix form in the diagonalised ϵ molecular orbital set,

$$F\phi_i = \epsilon_i \phi_i. \tag{23}$$

Here, ϵ_i is the Hartree–Fock orbital energy, and the associated ϕ_i describes the ith orbital of the atomic system. The hydrogen atom has analytic solutions to the Schrödinger equation and hence analytic orbitals [4]. More complex systems of interest do not generally have analytic solutions. As such, ϕ_i cannot be solved for analytically and must instead be approximated using a set of basis functions spanning a finite-dimensional subspace of Hilbert space [5]. Slater-type orbitals (STOs) use solved hydrogen orbitals as the basis functions:

$$\chi(\vec{r}) = f_{n-1}(r) r^{-\ell} P_\ell(x, y, z) e^{-r/n}.$$

Thijssen [6] gives the hydrogenic orbitals in this form. Here, ℓ is the angular-momentum quantum number, P_ℓ is a polynomial of degree ℓ, and f_{n-1} is a polynomial of degree $n-1$, where n is the principal quantum number. A general form of a hydrogenic orbital is chosen for STOs,

$$\chi_\zeta(\vec{r}) = r_A^m P_\ell(x, y, z) e^{-\zeta |\vec{r} - \vec{R}_A|}. \tag{24}$$

These basis functions are centred around a nucleus at location $\vec{R}_A$ relative to the origin. The parameter ζ defines the range of the orbital and can be determined by variationally optimising the Hartree–Fock energy [5]. The computational

difficulty with STOs is that the Hartree–Fock integrals are difficult to evaluate. The overlap matrix elements, for example, are defined by

$$S_{pq} = \langle \chi_p | \chi_q \rangle \, .$$

A basis set which simplifies the evaluation of these integrals was proposed in 1950 by Boys [5], who introduced Gaussian functions as basis functions:

$$\chi_\alpha(\vec{r}) = P_m(x, y, z) e^{-\alpha |\vec{r} - \vec{R}_A|^2}. \tag{25}$$

The polynomial P_m contains the angular component of the orbital. The angular dependence is given by the spherical harmonics $Y_\ell^m(\theta, \phi)$. For $\ell = 0$ (s-orbital), there is no angular dependence, and hence

$$\chi_\alpha^{(s)}(\vec{r}) = e^{-\alpha |\vec{r} - \vec{R}_A|^2}. \tag{26}$$

For $\ell = 1$, the three p-type Gaussian functions are

$$\begin{aligned} \chi_\alpha^{(p_x)}(\vec{r}) &= x e^{-\alpha |\vec{r} - \vec{R}_A|^2}, \\ \chi_\alpha^{(p_y)}(\vec{r}) &= y e^{-\alpha |\vec{r} - \vec{R}_A|^2}, \\ \chi_\alpha^{(p_z)}(\vec{r}) &= z e^{-\alpha |\vec{r} - \vec{R}_A|^2}. \end{aligned} \tag{27}$$

We are now in a position to calculate the orbitals of the atomic system of interest by expanding ϕ_i in Gaussian-type orbitals (GTOs):

$$\phi_i = \sum_\mu C_{\mu i} \chi_\mu.$$

Substituting this expansion into the Hartree–Fock equation (23) gives

$$F \sum_\mu C_{\mu i} \chi_\mu = \epsilon_i \sum_\mu C_{\mu i} \chi_\mu.$$

Projecting onto the basis function χ_η

$$\sum_\mu C_{\mu i} \underbrace{\langle \chi_\eta | F | \chi_\mu \rangle}_{F_{\eta\mu}} = \epsilon_i \sum_\mu C_{\mu i} \underbrace{\langle \chi_\eta | \chi_\mu \rangle}_{S_{\eta\mu}}.$$

In matrix notation,

$$FC_i = \epsilon_i S C_i.$$

For all orbitals,

$$FC = SC\epsilon. \tag{28}$$

He and Li^+ are described by ψ_{1s}, which I programmed as a combination of normalised s-type Gaussian wavefunctions, $\chi_\alpha^{(s)}(\vec{r}) = e^{-\alpha |\vec{r} - \vec{R}_A|^2}$ [7],

$$\psi_{1s} = \sum_i N_i C_i e^{-\zeta_i (\vec{r} - \vec{r_A})^2}.$$

The code constructs He and Li^+ separately, which holds true electronically for an infinite separation between He and Li^+. The calculation is simplified by setting the nucleus of each calculated system as the origin, so

$$\psi_{1s} = \sum_i C_i \underbrace{N_i e^{-\zeta_i r^2}}_{\chi_i}. \tag{29}$$

In the GitHub repository [7], the matrices F and S needed for the Hartree-Fock equation (28) are constructed in `functions_1s_atom.cpp`. These functions are called in `He_Li+_SCF.ipynb` and optimised with built-in Python linear algebra functions to build He and Li^+ out of Gaussians. Huzinaga [8] constructed He and $Li^+$with 10 Gaussian functions, enabling an informative comparison with our constructions. Firstly, in `functions_1s_atom.cpp` [7], a normalisation factor is constructed

$$\begin{aligned} \langle \chi_i | \chi_i \rangle &= 1, \\ N_i &= \left(\frac{2\zeta_i}{\pi} \right)^{3/4}. \end{aligned} \tag{30}$$

We can now build our matrices in C++ to solve eq. (28).

$$S_{ij} = \langle \chi_i | \chi_j \rangle = N_i N_j \left(\frac{\pi}{\zeta_i + \zeta_j} \right)^{3/2}. \tag{31}$$

$$\begin{aligned} h_{ij} = T_{ij} + V_{ij} &= \langle \chi_i \left| -\frac{1}{2}\nabla^2 \right| \chi_j \rangle + \langle \chi_i \left| -\frac{Z}{r} \right| \chi_j \rangle \\ &= \frac{3 N_i N_j \zeta_i \zeta_j \pi^{3/2}}{(\zeta_i + \zeta_j)^{5/2}} - \frac{2\pi Z N_i N_j}{\zeta_i + \zeta_j}. \end{aligned} \tag{32}$$

Here, $Z = 2$ for He and $Z = 3$ for Li^+. Furthermore, it can be shown [6] that the matrix elements of F are given by

$$F_{pq} = h_{pq} + \sum_k \sum_{rs} C^*_{rk} C_{sk} \left[\underbrace{2 \langle pr|g|qs \rangle}_{J_{pq}} - \underbrace{\langle pr|g|sq \rangle}_{K_{pq}} \right], \tag{33}$$

where $g = \frac{1}{|\vec{r}\,' - \vec{r}|}$. It can also be proven [6] that

$$\langle ij|g|kl \rangle = N_i N_j N_k N_l \frac{2\pi^{5/2}}{(\zeta_i + \zeta_k)(\zeta_j + \zeta_l)\sqrt{\zeta_i + \zeta_j + \zeta_k + \zeta_l}}. \tag{34}$$

In the C++ code, we define

$$G_{pq} = \sum_{rs} P_{rs} [\langle pr|g|qs \rangle - \frac{1}{2} \langle pr|g|sq \rangle], \tag{35}$$

Where P_{rs} is the density matrix,

$$P_{rs} = 2 \sum_{k} C_{rk} C_{sk}.$$

This form is chosen to allow for an optimisation procedure of the density matrix in `He_Li+_SCF.ipynb`.

Finally in the C++ code [7],

$$F_{pq} = h_{pq} + G_{pq}.$$

Thus, we have the matrices F and S necessary to solve eq. (28).

The Jupyter notebook `He_Li+_SCF.ipynb` [7] uses the functions of `functions_1s_atom.cpp` to build approximations for the $1s$ orbitals of He and Li^+ using ten Gaussian functions for each. In the Juypter notebook, the function

```
optimize_zetas(N, Z, n_electrons, convergence_threshold=1e-6, initial_zeta=None)
```

minimises the electronic energy of the system by changing the decay parameters ζ of the Gaussian basis functions. Minimisation utilises the Nelder–Mead simplex algorithm [9], as implemented by `scipy.optimize.minimize` in SciPy [10]. The objective being minimised is the electronic energy, and the variables changed to minimise the energy are the Gaussian decay constants ζ. After this optimisation procedure, the new density matrix and electronic energy are compared with the previous values to assess convergence.

Within the variational Hartree–Fock framework, a lower converged energy obtained using the same Hamiltonian represents an improved Hartree–Fock approximation [4]. The present results are compared with Huzinaga's Gaussian-basis calculations [8] that were used to construct He and Li^+ by Celli et al. [11]. The present values are also evaluated in reference to the lowest Hartree–Fock limits reported in the literature by Baskerville et al. [13].

For helium, the calculated energies in Hartree units are

$$E_{\text{present}} = -2.861672978372\, E_\text{h}, \qquad E_{\text{Huzinaga}} = -2.861669230222\, E_\text{h}, \qquad E_{\text{Baskerville}} = -2.861679995612\, E_\text{h}.$$

The present result is lower than Huzinaga's value by

$$E_{\text{present}} - E_{\text{Huzinaga}} = -3.748150 \times 10^{-6}\, E_\text{h},$$

and lies

$$E_{\text{present}} - E_{\text{HF}} = 7.017240 \times 10^{-6}\, E_\text{h}$$

above the Baskerville et al. limit.

Similarly, for Li^+,

$$E_{\text{present}} = -7.236395295401\, E_\text{h}, \qquad E_{\text{Huzinaga}} = -7.236307\, E_\text{h}, \qquad E_{\text{Baskerville}} = -7.236415201452\, E_\text{h}.$$

The present result improves upon Huzinaga's value by

$$E_{\text{present}} - E_{\text{Huzinaga}} = -8.829540 \times 10^{-5}\, E_{\text{h}},$$

with a remaining difference from the Baskerville et al. Hartree–Fock limit of

$$E_{\text{present}} - E_{\text{Baskerville}} = 1.990605 \times 10^{-5}\, E_{\text{h}}.$$

Thus, for both He and Li^{T}, the present calculations give variationally lower energies than the corresponding Huzinaga results which were used to approximate the Born-Mayer repulsive interaction between He and Li^{+} (more to be added here...) come back.

Equipped with the Gaussian function expansions of the $1s$ orbitals of He and Li^{+}, the Hartree–Fock interaction energy of the atom and ion can be evaluated at R, the internuclear distance between He and Li^{+},

$$E_{\text{int}}^{\text{HF}}(R) = E_{\text{He}-\text{Li}^{+}}(R) - E_{\text{He}} - E_{\text{Li}^{+}}. \tag{36}$$

E_{He} is the energy found for the isolated He atom. Following the logic of Celli et al. [11], first-order SCF energies are used to describe the repulsive potential between He and Li^{+}. The second-order induction and dispersion terms are added separately with damping functions [12]

$$\begin{aligned} V(R) = E_{\text{SCF}}^{(1)} + E_{\text{induction}}^{(2)} + E_{\text{dispersion}}^{(2)} + E_{\text{intra}} \\ + E_{\text{ex-disp}} + E_{\text{ex-ind}} + \text{higher-order terms}. \end{aligned} \tag{37}$$

At infinite separation, $R \to \infty$, He and Li^{+} are isolated and their $1s$ orbitals can be calculated independently. These isolated $1s$ orbitals are subsequently frozen and used unchanged at every finite internuclear separation R. This is an approximation, as the orbitals would in reality polarise and adjust from their spherical, isolated $1s$ configurations. The interaction at short-range separation is dominated by electron-cloud overlap and the associated exchange interaction of Equation (5). Atomic electron density decays approximately exponentially with distance [14]. Hence, the exchange repulsion arising from the Pauli exclusion principle can be approximated by an exponential decay with internuclear separation. This repulsion is known as the Born–Mayer repulsion [15].

When calculating the interacting system of He and Li^{+}, every new evaluation of R generates a unique atomic-orbital overlap matrix S, whose elements

$$S_{\mu\eta} = \langle \chi_\mu | \chi_\eta \rangle$$

describe the overlap between the individual Gaussian basis functions χ_μ and χ_η. For N Gaussian functions describing He and M describing Li^{+}, S is an $(N+M) \times (N+M)$ matrix.

The coefficients to describe the interacting He and Li^+ system, we initally call C_{guess}. Here we independetly combine the calculated Gaussian basis functions of isolated He and Li^+ to form an interacting system. The overlap between these two complete orbitals is therefore described by the 2×2 matrix O,

$$O_{ij} = \sum_{\mu\eta} C_{\mu i} S_{\mu\eta} C_{\eta j} = (C^T_{\text{guess}} S C_{\text{guess}})_{ij}. \tag{38}$$

While S contains the overlaps between every pair of Gaussian basis functions, O contains the overlaps between the two occupied orbitals constructed from them. At finite R, the off-diagonal elements of O represent the overlap between the frozen He and Li^+ $1s$ orbitals. O has the form

$$O = \begin{pmatrix} 1 & \langle \phi_{\text{He}} | \phi_{\text{Li}^+} \rangle \\ \langle \phi_{\text{Li}^+} | \phi_{\text{He}} \rangle & 1 \end{pmatrix}. \tag{39}$$

The frozen $1s$ orbitals are represented by Gaussian expansions,

$$\phi_{\text{He}}(\mathbf{r}) = \sum_{\mu=1}^{N} C^{\text{He}}_{\mu} N^{\text{He}}_{\mu} e^{-\alpha^{\text{He}}_{\mu} |\mathbf{r} - \mathbf{R}_{\text{He}}|^2}, \tag{40}$$

and

$$\phi_{\text{Li}^+}(\mathbf{r}) = \sum_{\eta=1}^{M} C^{\text{Li}^+}_{\eta} N^{\text{Li}^+}_{\eta} e^{-\alpha^{\text{Li}^+}_{\eta} |\mathbf{r} - \mathbf{R}_{\text{Li}^+}|^2}. \tag{41}$$

For the Hartree–Fock calculation, the occupied orbitals are transformed into an orthonormal set satisfying

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}, \tag{42}$$

such that their orbital-overlap matrix becomes

$$O = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \tag{43}$$

This ensures ensure that the two occupied orbitals are othonormal, a necessary condition for simplifying Roothan's equations for the SCF proecdure [1]. Firstly, the occupied-orbital overlap matrix O is diagonalised by `np.linalg.eigh()` [16],

$$OU = U\Lambda, \tag{44}$$

where the columns of U are the eigenvectors of O and Λ contains the corresponding eigenvalues. Importantly, the eigenvectors define linear combinations of the guessed orbitals that are mutually orthogonal. One can then transform the C coefficients of the interacting system to produce orthogonal orbitals,

$$C_{\text{orthogonal}} = C_{\text{guess}} U. \tag{45}$$

The eigenvalues in Λ give the squared norms of these orthogonal orbitals. Dividing each orbital by the square root of its corresponding eigenvalue gives the S-orthonormal occupied orbitals,

$$C_{\text{orthonormal}} = C_{\text{guess}} U \Lambda^{-1/2}, \tag{46}$$

such that

$$O = C_{\text{orthonormal}}^{T} S C_{\text{orthonormal}} = I. \tag{47}$$

This procedure, which provides S-orthonormal occupied orbitals for the He and Li^+ interaction, is implemented by the function `s_orthonormalise_occupied_guess` in `He_Li+_SCF.ipynb` [7]:

```
def s_orthonormalise_occupied_guess(C_guess, S):
    C_guess = np.array(C_guess, dtype=float)
    S = np.array(S, dtype=float)

    orbital_overlap = C_guess.T @ S @ C_guess

    overlap_values, orbital_combinations = np.linalg.eigh(
        orbital_overlap
    )

    orthogonal_orbitals = C_guess @ orbital_combinations

    normalisation = np.diag(1.0 / np.sqrt(overlap_values))
    orthonormal_orbitals = orthogonal_orbitals @ normalisation

    return orthonormal_orbitals
```

Once the C coefficients describing the interacting He and Li^+ system are orthonormalised, the electronic energy of the frozen-orbital determinant is evaluated using the closed-shell Hartree–Fock energy expression [6],

$$E_{\text{electronic}} = \frac{1}{2} \sum_{pq} P_{pq} \left(h_{pq} + F_{pq} \right). \tag{48}$$

Here, P_{pq} is the density matrix constructed from the occupied frozen orbitals, h_{pq} is the core-Hamiltonian matrix containing the electronic kinetic-energy and electron–nuclear-attraction contributions, and $F_{pq} = h_{pq} + G_{pq}$ is the Fock matrix. To construct the Fock matrix for basis functions on different nuclei, the molecular Gaussian integrals of Refs. [17, 18] are implemented in `Functions_1s_molecular.cpp` [7] and utilised by `He_Li+_SCF.ipynb`. Consider two normalised s-type Gaussian basis functions,

$$\chi_a(\mathbf{r}) = N_a e^{-\zeta_a |\mathbf{r} - \mathbf{A}_a|^2},$$

and

$$\chi_b(\mathbf{r}) = N_b e^{-\zeta_b|\mathbf{r}-\mathbf{A}_b|^2}.$$

Define

$$p_{ab} = \zeta_a + \zeta_b, \qquad \mu_{ab} = \frac{\zeta_a\zeta_b}{p_{ab}}, \qquad \mathbf{P}_{ab} = \frac{\zeta_a\mathbf{A}_a + \zeta_b\mathbf{A}_b}{p_{ab}}.$$

The molecular integrals implemented in `Functions_1s_molecular.cpp` are

$$\begin{aligned}
S_{ij} &= N_i N_j \left(\frac{\pi}{p_{ij}}\right)^{3/2} e^{-\mu_{ij}A_{ij}^2}, \\
T_{ij} &= S_{ij}\mu_{ij}\left(3 - 2\mu_{ij}A_{ij}^2\right), \\
V_{ij} &= -N_i N_j \frac{2\pi}{p_{ij}} e^{-\mu_{ij}A_{ij}^2} \sum_C Z_C F_0\left(p_{ij}|\mathbf{P}_{ij} - \mathbf{C}|^2\right), \\
I_{ijkl} &= \frac{2\pi^{5/2} N_i N_j N_k N_l}{p_{ik}p_{jl}\sqrt{p_{ik}+p_{jl}}} e^{-\mu_{ik}A_{ik}^2 - \mu_{jl}A_{jl}^2} F_0\left(\frac{p_{ik}p_{jl}}{p_{ik}+p_{jl}}|\mathbf{P}_{ik} - \mathbf{P}_{jl}|^2\right),
\end{aligned} \tag{49}$$

where $A_{ab} = |\mathbf{A}_a - \mathbf{A}_b|$ and

$$F_0(t) = \frac{\sqrt{\pi}}{2\sqrt{t}}\,\mathrm{erf}(\sqrt{t}), \qquad F_0(0) = 1.$$

The code takes into consideration the possible centre assignments of χ_α and χ_β. These consist of both orbitals being centred on He, both being centred on Li^+, or one being centred on each nucleus. `He_Li+_SCF.ipynb` places the He nucleus, $\vec{A}_a$, at the origin and the Li^+ nucleus, $\vec{A}_b$, at the internuclear displacement $\vec{R}$. Once `He_Li+_SCF.ipynb` has evaluated the molecular $He - Li^+$ electronic energy, the nuclear repulsion is added:

$$E_{\mathrm{total}}(R) = \underbrace{\frac{1}{2}\sum_{pq} P_{pq}(h_{pq} + F_{pq})}_{E_{\mathrm{electronic}}} + \underbrace{\frac{Z_{\mathrm{He}}Z_{\mathrm{Li}}}{R}}_{\text{nuclear repulsion}}. \tag{50}$$

Finally, the repulsive interaction energy is calculated in `He_Li+_SCF.ipynb` using

$$V_{\mathrm{repulsion}}(R) = E_{\mathrm{total}}(R) - E_{\mathrm{He}}^{\mathrm{isolated}} - E_{\mathrm{Li^+}}^{\mathrm{isolated}}, \tag{51}$$

by iterating through R from 6.0 to 1.0 bohr in intervals of 0.1 bohr. The positive interaction energies between 1.5 and 3.2 bohr are then fitted to the Born–Mayer form

$$V_{\mathrm{repulsion}}(R) = Ae^{-bR}. \tag{52}$$

For the chosen fitting interval, the present calculation gives

$$A_{\mathrm{present}} = 22.2644\,E_{\mathrm{h}}, \qquad b_{\mathrm{present}} = 2.66712\,a_0^{-1}. \tag{53}$$

Ahlrichs et al. reported $A = 23.92\,E_{\mathrm{h}}$ and $b = 2.694\,a_0^{-1}$ for the $He - Li^+$ Born–Mayer repulsion [12]. The present calculation employs a Gaussian basis set that

gives lower isolated-atom Hartree–Fock energies than the Huzinaga expansions used by Ahlrichs et al. [8]. Within the variational Hartree–Fock framework [4], these lower energies indicate improved representations of the isolated orbitals. Since these orbitals form the basis of the frozen-orbital interaction calculation, the present treatment is expected to provide a more accurate first-order repulsive interaction and, consequently, improved Born–Mayer parameters.

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